Research Square

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Pyridostigmine and low-dose naltrexone for ME/CFS: study protocol for the Life Improvement Trial (LIFT), a randomized, double-blind, placebo-controlled clinical trial

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Research Article

Keywords: ME/CFS, LDN, pyridostigmine, Mestinon, FUNCAP-55, Garmin, PROMIS-29, DSQ-PEM, SHAPE

Posted Date: March 4th, 2025

DOI: https://doi.org/10.21203/rs.3.rs-5626167/v1

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Version of Record: A version of this preprint was published at Trials on August 12th, 2026. See the published version at https://doi.org/10.1186/s13063-026-09943-6.

Abstract

Background

Myalgic encephalomyelitis/chronic fatigue syndrome (ME/CFS) is a complex, chronic disease with no FDA-approved treatments. This report describes a protocol for the Life Improvement Trial (LIFT), a randomized, double-blind, placebo-controlled clinical trial investigating the impact of low-dose naltrexone (LDN) and pyridostigmine (Mestinon) on physiological response, symptoms, and functionality of ME/CFS patients.

Methods

Participants (target n = 160) are recruited through clinics at Massachusetts General Hospital and Brigham and Women's Hospital, and through Open Medicine Foundation's StudyME registry. They are then randomized into one of four arms: LDN/pyridostigmine, LDN/placebo, placebo/pyridostigmine, placebo/placebo. Treatment is administered for 13 weeks after an initial screening period of up to 4 weeks. Primary outcomes are FUNCAP-55 score, peak oxygen utilization, heart rate recovery, and oxygen uptake efficiency slope. Secondary outcomes are scores from DSQ-PEM and PROMIS-29 surveys, DANA Brain Vital score, step count, heart rate, and heart rate variability.

Discussion

The results of this trial will provide novel insights into the efficacy of and predictors of response to LDN and pyridostigmine in ME/CFS. This may inform future treatment strategies for ME/CFS. The trial will also validate what primary and secondary outcomes to use in similar clinical trials.

Trial registration:

This trial was registered on clinicaltrials.gov (NCT06366724) on April 16, 2024.

Background

Myalgic encephalomyelitis/chronic fatigue syndrome (ME/CFS) is a chronic, debilitating, complex disease that impacts multiple systems in the body, including the immune, metabolic, cardiovascular, and nervous systems. (1) Its diagnosis relies on identifying hallmark symptoms such as profound fatigue, unrefreshing sleep, orthostatic intolerance, neurocognitive disturbances and brain fog, pain, and post-exertional malaise (PEM)—the worsening of symptoms following even mild physical or cognitive exertion—to identify patients. (2) There are no FDA-approved treatments for ME/CFS, so clinical trials to identify efficacious interventions are critically needed. (1)

Pyridostigmine, or Mestinon, and low-dose naltrexone (LDN) are two drugs that have shown promise in the treatment of ME/CFS. Pyridostigmine is an acetylcholinesterase inhibitor, which can increase the level and duration of acetylcholine in the neural synapse. (3,4) It is currently FDA-approved for the treatment of myasthenia gravis. (5) Naltrexone is an approved treatment for alcohol and/or opioid use disorder, which has anti-inflammatory effects at low doses (0.5–4.5 mg), making LDN relevant for the treatment of ME/CFS. (6) Pyridostigmine and LDN address different underpinnings of symptoms in ME/CFS: neurovascular dysregulation and inflammation, respectively. The combination of the two drugs has never been studied.

People with ME/CFS often exhibit vascular dysregulation, with low biventricular filling pressures (preload failure) resulting in decreased aerobic capacity. (7) By increasing levels of acetylcholine, it is thought that pyridostigmine increases cholinergic stimulation of norepinephrine release at the post-ganglionic synapse, which, in turn, improves constriction of capacitance vessels and return of blood to the heart. This may then lead to improved right atrial pressure in ME/CFS patients. (8)

Previous work has demonstrated that pyridostigmine can improve aerobic capacity in ME/CFS. (9) As measured by a sub-maximum exercise heart and pulmonary evaluation (SHAPE) test, pyridostigmine treatment increases oxygen uptake efficiency and pulmonary vascular capacitance. (9) In postural orthostatic tachycardia syndrome (POTS), a common comorbidity of ME/CFS, pyridostigmine has been shown to reduce standing heart rate, improve diastolic blood pressure, and significantly decrease symptom burden after acute use. (10,11) Acetylcholinesterase inhibitors are also used to improve patients' orthostatic intolerance, which is an important symptom of ME/CFS. (12) There is limited evidence on the long-term efficacy of pyridostigmine, however, so structured clinical trials are required.

In addition to vascular dysregulation and orthostatic intolerance, chronic inflammation is a common occurrence in ME/CFS. (6) LDN is thought to address this by modulating glial cells and peripheral blood mononuclear cells as a toll-like receptor (TLR) antagonist. (6,13–16) Through TLR-4 regulation and modulation of the response to intracellular TLR activity, LDN reduces pro-inflammatory cytokines like interleukin-1, interleukin-6, tumor necrosis factor-alpha, interferon beta, and nitric oxide. (6,13) This anti-inflammatory effect thus reduces fatigue, inflammation, pain, and cognitive issues. (6,14,16)

In the clinic, LDN treatment has produced positive responses in people with ME/CFS (17) and similar diseases, including fibromyalgia (18), multiple sclerosis, irritable bowel syndrome, and Crohn's disease. (6,19,20) A retrospective analysis of 218 ME/CFS patients in Finland found that 73.9% of people experienced less pain and improved physical and cognitive performance after LDN treatment. (19) In Long COVID—a disease with overlapping symptomology and potentially pathophysiology as ME/CFS (21)—a 2-month treatment period with LDN was shown to reduce symptom burden, including pain levels, sleep disturbance, and limitation in daily living activities. (22) The majority of prospective evidence on the efficacy of LDN in ME/CFS and related conditions is subjective with limited sample sizes, however, so large trials with objective outcomes are necessary.

This clinical trial, designated as the Life Improvement Trial (LIFT), aims to evaluate the effect of pyridostigmine and LDN on the functional capacity, physiological response, and symptoms of ME/CFS via a randomized, double-blind, placebo-controlled study design. This is the first known investigation of potential additive effects of using the two drugs in combination. As there are no approved treatments for ME/CFS or standard of care, the trial aims to show superiority over placebo, the chosen comparator. The trial will also evaluate any potential differences between responders and non-responders. Ultimately, the LIFT will provide rigorous scientific evidence of the clinical utility of pyridostigmine and LDN, which may lead to more personalized treatment plans for people with ME/CFS in the future.

Methods/Design

This report follows SPIRIT reporting guidelines. (23) A completed SPIRIT checklist is provided as supplementary material.

Trial design overview

The LIFT is a randomized, double-blind, placebo-controlled clinical trial. The study was approved by the Institutional Review Board of Massachusetts General Brigham (2024P000087). In-person recruitment is through clinics at Massachusetts General Hospital and Brigham and Women's Hospital. Further recruitment is through Open Medicine Foundation's StudyME registry.

Participants are randomized (1:1:1:1) to placebo/placebo, placebo/LDN, pyridostigmine/placebo, and pyridostigmine/LDN arms. Randomization is conducted by the Brigham and Women's Hospital Investigational Drug Services (IDS), and arm assignment is not shared with trial personnel or participants. Randomization is stratified by sex (male/female), and the randomization code will not be released by the pharmacy until the completion of the trial to ensure allocation concealment. As a result, all assessments will be completed by staff blind to treatment allocation. The full study schema is shown in Fig. 1.

All participants receive a Garmin Vivosmart® 5 watch at no cost. Data from Garmin watches is collected through Labfront (24). Participant-reported outcomes (i.e., surveys) are completed through the REDCap web application. All participants also use the DANA Modular mobile application (25) to complete Brain Vital cognitive assessments. See Fig. 2 for a full overview of data collection throughout the treatment period.

Participants attend three in-person visits at the Clinical Trials Hub at Brigham and Women's Hospital: for screening, to initiate the treatment period, and at the conclusion of the study 13 weeks later. A SHAPE test (26–28) and biosample collection (blood draw and urine) are performed during the two latter in-person visits. Blood samples are frozen and stored at -80°C for future analyses, including proteomics and metabolomics, that might be used to determine if there are measurable differences between responders and non-responders. Coordinators also conduct periodic virtual visits with participants throughout the treatment period. For the full study schedule, see Fig. 3.

Patient representative involvement

Four patient representatives were asked to provide feedback on the LIFT protocol. Their review, in addition to input from the larger patient community, directly resulted in protocol updates, including expanding the window for completing questionnaires, the timeline for post-treatment follow-up, and changes to the dosage titration schedule.

Participants

Adults aged 18–65 may be eligible to participate in the LIFT if they meet the Canadian consensus criteria for ME/CFS and onset occurred prior to 2020. See Table 1 for full eligibility criteria. People who are treatment naïve are prioritized. Participants are also asked to withhold the following medications for at least one month before beginning the study: “mitochondrial cocktail” (including coenzyme Q_{10}, carnitine, creatine, riboflavin, alpha lipoic acid), midodrine, fludrocortisone, intravenous immunoglobulin, and Hizentra.

Sample size

With a sample size of 40 per trial arm (160 total participants), preliminary power calculations indicate the ability to detect moderate and large effects from the two interventions under reasonable model assumptions. (29,30) However, the trial may not be powered to detect the synergistic interaction between the two medications, particularly after adjusting for multiple comparisons. To account for participant attrition, the trial will target enrollment of 180 patients.

Recruitment, screening, and consent

Participants are recruited through two primary methods. Patients may be recruited through Dr. David Systrom’s or Dr. Donna Felsenstein’s clinics at Brigham and Women’s Hospital or Massachusetts General Hospital, respectively. Alternatively, potential participants may be recruited through the Open Medicine Foundation StudyME registry. Subjects recruited through StudyME are asked to provide access to their medical records for an initial prescreen. Medical records for all participants are used for inclusion/exclusion purposes, including screening for prior LDN and pyridostigmine use.

Potential participants are further screened during a visit to the Clinical Trials Hub at Brigham and Women’s Hospital. In addition to answering questions about the person’s medical history and signing the consent form, during the visit, research coordinators measure the subject’s height and weight, take vital signs, collect a urine sample to test for pregnancy and drugs of abuse, and set the subject up with a Garmin watch and LabFront and DANA Brain Vital accounts. Each participant is informed about their study status after lab results are returned and assessed by study staff, usually within 72 hours.

Participants are provided with the informed consent form during the screening period, up to 28 days prior to their study visit initiating treatment. At the end of the screening period, coordinators contact the participant to confirm eligibility and answer questions about the consent form.

Participant compensation

Participants are compensated for their time and effort via an Advarra payment card, which is based on activities completed:

- Screening visit (in person): $50
- Visit 1 (in person): $75
- Visit 2/3/4 (virtual): $25
- Visit 5 (in person): $75
- Visit 6 (virtual): $25

Subjects who complete the trial are also permitted to keep their Garmin watch.

Interventions

LDN and pyridostigmine can be administered in many ways, with the most commonly used formulations being liquid and tablets. (31,32) Both formulations are used clinically, though anecdotal reports from clinicians have indicated the liquid formulation is often better tolerated by patients. For this reason and to better allow up and down titration of drugs and placebo, this trial uses liquid formulations.

LDN and pyridostigmine are titrated up to their target doses over a period of 4 weeks. LDN is administered at 1.5 mg per day for the first 2 weeks and 3 mg per day for the following 2 weeks before reaching 4.5 mg per day for the remainder of the treatment period. Pyridostigmine follows a similar pattern, increasing from 20 mg three times per day to 40 mg three times per day to 60 mg three times per day (180 mg total daily) in two-week intervals.

If a participant reports an adverse event or that they are not tolerating their current dose, the dose will be reduced to the previous level. See the full dosing schedule with intolerance plan in Fig. 4.

Compliance monitoring

Participant compliance is monitored by coordinators via study visits, both in person and virtual. In addition, participants are asked to maintain a dosing diary throughout the treatment period, noting when they complete their daily doses. Missing a single dose is recorded as a minor deviation. Missing several doses sequentially is considered a major deviation and is reported to the IRB.

Adverse event monitoring

Coordinators perform an adverse event assessment during each in-person and virtual visit, taking place on weeks 2, 4, 10, 13, and two weeks following the treatment phase for follow-up. Participants are also given the principal investigator's contact information so they can discuss anything they experience in between scheduled visits. If an adverse event is deemed severe and causally linked to the intervention, the participant may be unblinded before the completion of the trial.

Outcome measures

The primary outcomes of the trial are FUNCAP-55 score, peak oxygen utilization, heart rate recovery, and oxygen uptake efficiency slope. Secondary outcome measures include cognitive testing, wearable device data, and symptom severity, as measured via surveys. FUNCAP-55 (33), PROMIS-29 (34), and DSQ-PEM (35) surveys are all validated measures. See Table 2 for the complete list of primary and secondary outcomes.

Members of Open Medicine Foundation's computation team will monitor data quality periodically throughout the trial. Data will be stored on secure servers, and blood samples banked for future analyses will be identified by a coded ID number to preserve participant confidentiality.

Statistical analysis

Mixed effects models are a powerful way to model data with multiple timepoints and subgroups, as is the case with the LIFT. Given a mixed model approach, p-values for this trial will be calculated via the Satterthwaite's degrees of freedom method or the Kenwood-Roger approximation.

The testing for each hypothesis will determine whether the change in the given variable across the time points is significantly different in the three case groups, relative to the controls. The significance level will be set at $p < 0.05$ and adjusted for multiple tests by controlling the family-wise error rate across the primary tests (three groups, relative to the control group, with four outcomes for each). The secondary outcomes will be tested in the same manner, accounting additionally for the primary tests. Statistical analyses will incorporate age and BMI as covariates, and the model will be run first as a whole model and then partitioned into male/female groups if appropriately balanced.

A further secondary analysis will use classification techniques to identify the principal variables influencing responder/non-responder status to the intervention among the participants. Such variables will be drawn from the baseline information collected at the initial visit.

Discussion

This report details the protocol for a 13-week randomized, double-blind, placebo-controlled clinical trial evaluating the efficacy of LDN and pyridostigmine in a pre-COVID-19 pandemic ME/CFS population. The study gathers physiological and symptom-based evidence of the impact of both drugs and will enable the investigation of potential predictors of response to each drug.

Limitations

While the trial is preferentially enrolling treatment naïve participants, the limited availability of such patients may require that some participants will be enrolled in the study having previous experience with one or both medications. In such an event, it is possible that the blinding for these participants may be compromised if they either respond or do not respond as they did in their previous experience. This risk is partially mitigated by the formulation of the drugs in the trial. The LDN, pyridostigmine, and placebo solutions are flavored to mask the taste of either of the drugs, which reduces the potential risk of identifying study arm assignment based on previous experience.

The lack of standardized outcome measures for ME/CFS research limits the ability to compare efficacy results from this trial with other studies. The primary outcomes for the LIFT were selected as objective measures of physiological response relevant to the potential mechanism of action of the drugs, as well as quasi-objective, validated survey measures of symptoms. Secondary outcomes include surveys that are frequently used in other research studies to facilitate comparisons where possible.

The use of the SHAPE test necessitates that participants are able to travel to the clinic, which excludes housebound and bedbound patients. Therefore, trial participants may differ from non-participants who otherwise fit the study inclusion criteria. Randomization mitigates the impact of a heterogeneous patient population, but the trial results may not be generalizable to ME/CFS patients of all severity levels.

Significance

The LIFT constitutes a significant contribution to the ME/CFS research field, with large potential implications for the future of clinical management of the disease. ME/CFS has no FDA-approved treatments. The LIFT is a randomized, double-blind, placebo-controlled trial, and as one of the largest trials ever conducted in the field, it is a major step forward towards providing patients with recognized, approved clinical treatment.

It is widely recognized that the ME/CFS patient population is heterogeneous, where each patient reacts differently to the same treatment strategy. This clinical trial acknowledges the heterogeneous nature of the patients and works to identify differences in responders and non-responders. Identifying these differences has the potential to transform clinical practices, enabling a precision medicine approach to treatment strategies and ultimately improve outcomes.

Trial status

Protocol version 04, June 24, 2024. Further modifications to the protocol that impact how the study is conducted will require an amendment, which will be approved by the Institutional Review Board prior to implementation. Recruitment began in August, 2024 and is planned to be completed by May, 2026.

Abbreviations

LDN: low-dose naltrexone

LIFT: Life Improvement Trial

ME/CFS: myalgic encephalomyelitis/chronic fatigue syndrome

SHAPE: sub-maximum exercise heart and pulmonary evaluation

Declarations

Ethics approval and consent to participate

Institutional Review Board of Massachusetts General Brigham (2024P000087). Written informed consent to participate will be obtained from all participants by clinical research coordinators.

Consent for publication

Not applicable

Availability of data and materials

Trial results will be published in an open access journal following completion of the study. The final trial dataset will be de-identified and made available in the public domain through an open access repository.

Competing interests

LT is the CEO and founder of Open Medicine Foundation (OMF), and DM is employed by OMF. DS is the Director of OMF's Ronald G. Tompkins Harvard ME/CFS Collaboration. WX is the Director of OMF's Computational Research Center for Complex Diseases. JB is the Director of OMF's Collaborative Center at Uppsala and serves as OMF's Chief Medical Officer.

Funding

This work is fully funded by Open Medicine Foundation. Employees of the Foundation were involved in the design of the study and in writing the manuscript.

Authors' contributions

Conceptualization: DS, LT, JB, WX

Trial design: JS, JD, DM, LT, DS, PL, JB, WX, DF

Acquisition, analysis, or interpretation of data: JS, JD, SP, DF, PL, WX, DS

Drafting of manuscript: DM

All authors read and approved the final manuscript.

Acknowledgements

Not applicable

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Tables

Tables are available in the Supplementary Files section.

Figures

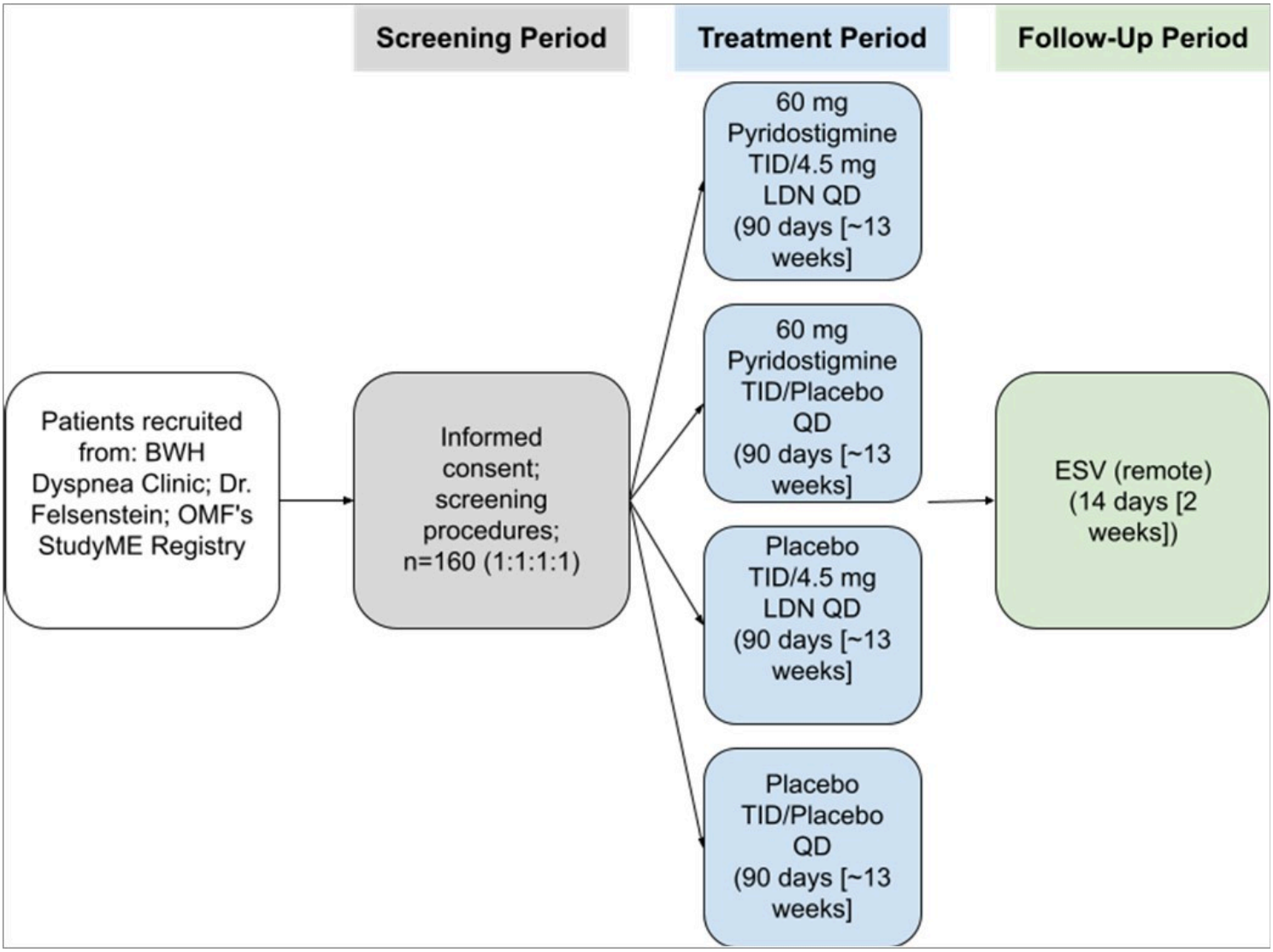


Figure 1

Study schema

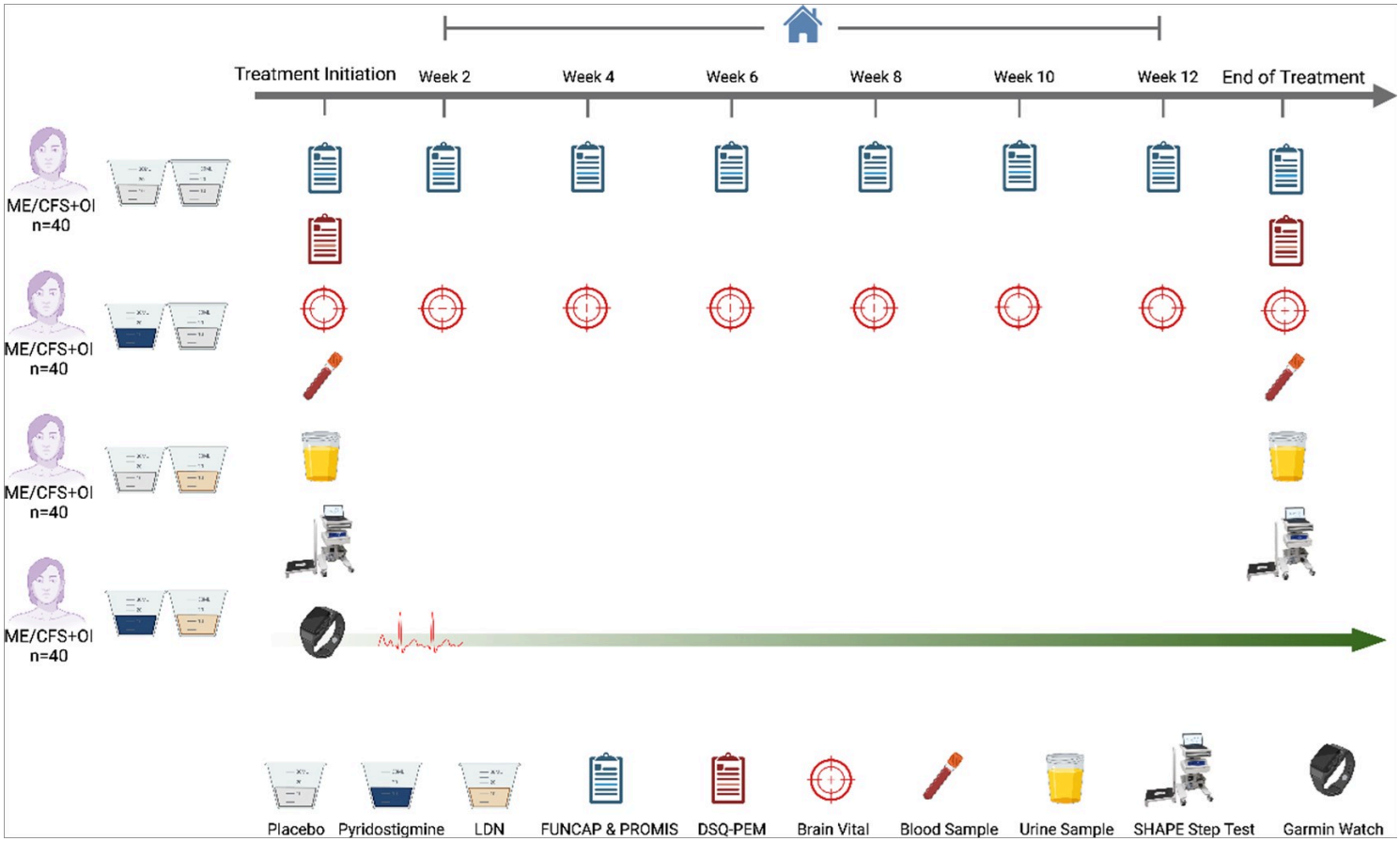


Figure 2

Overview of data collection during treatment period (created in BioRender.com)

	Screening Period (Days)	Treatment Period (Days)								Follow-Up Period
	-28 to -1	0	14	28	42	56	70	84	90	104
Enrollment										
Eligibility screen	X	X								
Informed consent	X									
Randomization		X								
Interventions										
Dispense/Collect IP		X							X	
IP Administration		→	→	→	→	→	→	→	→	
Assessments										
Demographics	X									
Medical History	X									
Height, Weight, Vitals	X	X							X	
SHAPE Test		X							X	
FUNCAP-55		X	X	X	X	X	X	X	X	
DSQ-PEM		X							X	
PROMIS-29		X	X	X	X	X	X	X	X	
Blood Draw		X							X	
Collect Garmin Data	→									→
Brain Vital	X	X	X	X	X	X	X	X	X	X

Figure 3

Schedule of enrollment, interventions, and assessments



Figure 4

Titration schedule during treatment period, including intolerance plan (created in BioRender.com)

Supplementary Files

This is a list of supplementary files associated with this preprint. Click to download.

- SPIRITchecklistFinal.docx
- Tables.docx